

Information Geometry & Dynamical Systems: Literature and Framework Mapping

Key Literature Threads

Information Geometry (IG) – Classical and Modern: IG studies statistical manifolds of probability distributions using tools like the Fisher information metric and dual affine connections (exponential and mixture connections) ¹. This field was pioneered by Shun-ichi Amari and others, building on Čencov's theorem that the Fisher metric is essentially the unique Riemannian metric invariant under sufficient-statistic (Markov) mappings ². Amari's α -geometry formalism shows that any Csiszár f -divergence on a probability manifold induces a Riemannian structure (the Fisher metric) and a pair of dual $\pm\alpha$ affine connections ¹. In particular, the one-parameter family of α -divergences yields a family of dual connections interpolating between the exponential and mixture families. Modern IG research extends these ideas to broader settings (e.g. quantum states, optimal transport) and emphasizes the geometric understanding of inference, model complexity, and information loss ³ ².

Operator-Theoretic Dynamics – Koopman, Perron–Frobenius, Semigroups: In dynamical systems theory, an alternative to state-space analysis is provided by linear operators that evolve functions or measures. The *Koopman operator* (introduced by B. O. Koopman in 1931) acts on observables (functions of state), advancing them through the dynamics, while the *Perron–Frobenius operator* acts on probability densities ⁴. These two are adjoint (dual) to each other on appropriate function spaces ⁴. The remarkable feature of this approach is that it represents nonlinear dynamics as an (infinite-dimensional) linear evolution, enabling the use of spectral analysis and linear algebraic techniques ⁵. In fact, *all* information about a deterministic dynamical system is contained in the Koopman operator; no information is lost by lifting to this linear framework ⁵. Over the past decade, Koopman's spectral theory has been revitalized ⁶, with proofs of Koopman spectral decompositions and new concepts like *Koopman modes* that link eigenfunctions to physical flow features ⁷. The evolution operators form one-parameter semigroups $T(t)$ (for continuous-time flows) or iterated maps (for discrete time); the *Hille–Yosida theorem* characterizes the generators of such C_0 -semigroups ⁸ and is fundamental in connecting spectral properties of the generator (e.g. eigenvalues) to the long-term dynamics. Markov processes in particular are analyzed via their forward generators and semigroups (Kolmogorov or Frobenius–Perron operators) – a context where the Hille–Yosida conditions often apply ⁹. Overall, the operator-theoretic viewpoint (sometimes called “Koopmanism” ¹⁰) has provided a unifying language for stability, ergodic properties, and resonance structures of dynamics ¹¹, complementary to traditional phase-space methods.

Delay-Coordinate Embeddings and State-Space Reconstruction: Delay embedding theory addresses how to reconstruct an equivalent state space from time-series observations. *Takens' Theorem* (1981) showed that, for a generic smooth dynamical system with a d -dimensional attractor, the map that takes a single scalar time series $y(t)$ to a k -dimensional delay vector $(y(t), y(t+\tau), \dots, y(t+(k-1)\tau))$ is an *embedding* (a diffeomorphism onto the true state space) provided $k > 2d$ ¹² ¹³. In essence, a sufficiently large number of delays can capture the underlying system's geometry and allow one to *reconstruct the attractor* up to smooth coordinate changes ¹⁴ ¹². This was extended by Sauer, Yorke, and Casdagli to include

fractal attractors (requiring $k > 2 \cdot \dim_{\text{Lyapunov}}$) and has become a standard tool in nonlinear time-series analysis. In modern data-driven dynamics, delay embeddings are also used to enrich the feature space for linear model approximations. For example, *Dynamic Mode Decomposition (DMD)* and its extensions assume one has a sequence of state snapshots and seek a best-fit linear operator; including time-delayed coordinates (stacking Hankel matrices of time-series data) allows DMD to capture more complex spectral content. **Extended DMD (EDMD)** in particular is a popular algorithm that, given a dictionary of observables, finds a finite-dimensional approximation to the Koopman operator via least-squares regression ¹⁵. By augmenting the state with delay coordinates (or nonlinear basis functions), EDMD can approximate leading Koopman eigenvalues and modes. Notably, it has been proven that applying DMD to a *Hankel matrix* of delay-embedded data can, in the infinite-data limit, recover the true Koopman eigenfunctions and eigenvalues of an ergodic system ¹⁶. This connection (sometimes called Hankel-DMD or extended DMD) bridges classical embedding theorems with operator theory: delay coordinates make hidden spectral structure observable ¹⁶. In practice, delay embeddings are a special case of a broader idea of choosing an *observable embedding* that makes a system's dynamics approximately linear and thus amenable to Koopman spectral analysis ¹⁷ ¹⁸.

Interfaces of IG and Dynamics: Although information geometry was initially developed for static statistical models, there have been various efforts to apply IG to dynamic systems and processes. One thread is the IG of *time-series models* in statistics and signal processing. Amari himself in the 1980s explored a differential-geometric treatment of linear dynamical systems: for example, he introduced a Riemannian metric and dual connections on the manifold of transfer functions of invertible linear systems ¹⁹, and studied ARMA (autoregressive moving-average) time-series models as points on a statistical manifold (where the power spectral density provides a natural parameterization) ²⁰. This line of work treated identification of linear systems using IG, viewing system parameters through the lens of an exponential family (for AR models) or a mixture family (for MA models) ²¹. Subsequently, other researchers like Ohara, Tanaka, and Fujimoto (1990s) extended IG to control systems and signal processing, defining Fisher metrics on state-space model families and studying natural gradients for adaptive filters. Another thread is *Markov chain information geometry*: the state transition probabilities of a Markov process can be seen as an exponential family (if we exponential-tilt a base transition matrix) ²². Information-geometric approaches have been used to analyze the space of transition matrices and even the space of trajectories. For instance, Amari (2001) investigated the *hierarchical manifold of joint distributions* of a Markov chain of increasing length ²³. He introduced an “orthogonal” decomposition of dependencies (pairwise, triplewise, etc.) in a higher-order Markov chain by projecting the full joint distribution onto submanifolds representing lower-order dependencies ²⁴. In general, the *hierarchy of probability distributions* (marginals, conditionals, joints at multiple time steps) can be given a differential-geometric structure ²⁵. Recent work continues to merge IG with dynamics: e.g. studying gradient flows on the space of distributions (flowing along Fisher–Rao geodesics or dual affine directions) to understand nonequilibrium processes ²⁶ ²⁷. The space of probability densities on a graph or manifold can carry dual-flat IG structures (one on densities, one on fluxes) to describe dynamical evolutions such as population flows or chemical reaction kinetics ²⁶ ²⁷. In summary, there is a growing “IG of dynamics” literature, but it remains less consolidated than static IG ²⁸. Notable contributions include applying IG to: Fokker–Planck equations and diffusion processes (treating them as gradient flows in the space of distributions), analyzing *Markov processes via IG projections* (e.g. information projection of a joint distribution onto a family of product distributions to quantify independence at various time scales) ²⁵, and using IG in control and filtering theory (e.g. interpreting filtering as an IG projection in belief-space) ²⁹. The user’s program builds directly on this intersection, marrying IG concepts (divergences, metrics, connections) with operator dynamics (semigroups, spectra, embeddings).

Prior Work on Specific Concepts

- **Realization Spaces from Edge Marginals:** The user's framework starts from an *empirical edge marginal* $\Gamma_Y \in \mathcal{P}(Y \times Y)$, which is essentially the joint distribution of consecutive observations (Y_t, Y_{t+1}) . In a stationary time series, Γ_Y is the one-step transition distribution of the observed process. The idea of characterizing all possible *realizations* (hidden-state dynamical models) consistent with a given observed transition distribution is well-studied in system identification and hidden Markov model (HMM) theory. This is essentially an **inverse problem**: given the joint distribution of observables over one time-step (or over all finite sequences), find all underlying state-space models that could produce it ³⁰. In HMM terminology, two different state-space models are *observation-equivalent* if they induce the same distribution on observable sequences. Prior work has shown that if one restricts to *minimal* realizations (those with the smallest state dimension), the equivalence class is trivial: any two minimal HMMs that produce the same observable distribution must be related by a relabeling (permutation) of states ³¹. However, non-minimal realizations (with redundant states) form families parametrized by non-unique factorizations of the probability transition matrix. Indeed, describing the set of all HMMs consistent with given sequence probabilities is difficult in general ³⁰. A related result in realization theory is that this problem can be reduced to a **nonnegative matrix factorization**: given the matrix of joint probabilities of (Y_t, Y_{t+1}) (which Γ_Y represents), one seeks to factor it as $\Lambda \Phi$ (initial-state distribution times observation matrix) or ΦT (observation matrix times transition matrix) in all possible ways ³². Each factorization corresponds to a hidden state representation. Finding the minimal inner dimension of such a factorization is essentially finding the minimal number of states (the “rank” of the stochastic matrix in the probabilistic sense), which is known to be hard in general (and related to the nonnegative rank of a matrix ³²). In summary, the notion of a *realization space* $\mathcal{R}(\Gamma_Y)$ collecting all dynamical models consistent with an empirical edge marginal is well-grounded: it connects to classical HMM equivalence and minimal realization theory (e.g. the work of Blackwell, and Vanluyten et al. 2007 on HMM equivalence ³³). What is *not* as established is doing this within an **information-geometric** framework – that is a novel twist by the user.

- **IG-Equivalence of Realizations:** If two different realizations (X, Γ_X, h) and (Z, Γ_Z, r) induce the *same* Γ_Y on observables (perhaps exactly, or at least yield $D_f(\Gamma_Y | \Gamma_Y) = 0$ trivially), we might call them *IG-equivalent* in the sense that no f-divergence can distinguish the observable statistics they produce. In classical terms this is just *statistical equivalence* (*observational indistinguishability*). Prior work outside IG has characterized such cases: e.g., in HMMs, different state dynamics or emission mappings can give identical observable probabilities (leading to non-identifiability without further assumptions). The IG view would treat all such realizations as lying on the same *manifold of distributions* (perhaps an “IG leaf” as the user says). Amari's work on hierarchical models is relevant here: for example, Amari (2001) considered the manifold of all joint distributions with a given set of lower-order marginals ²⁴. Two models that share the same one-step marginal Γ_Y but differ in higher-order dependence structure would lie in the same *fiber* when one considers the projection from the full joint distribution to the one-step distribution. In IG terms, a *Markov morphism* (like a many-to-one mapping from a latent space X to the observed Y) is an *information loss* mapping, and Čencov's theorem implies that such mappings contract the Fisher metric ². If two realization models yield the same Γ_Y , they are *information-projectable* to the same point in the manifold $\mathcal{P}(Y \times Y)$. There isn't a specific prior literature on “IG-equivalence” per se, but the concept closely aligns with existing ideas of

observational equivalence in system theory and **Markov invariance** in information geometry (the fact that Fisher information cannot increase under a Markov coarse-graining of data ³⁴). In short, identifying and characterizing all models that satisfy $D_f(\Gamma_Y \parallel \Gamma_Z) = 0$ (for all or some f) is a new formulation, but it builds on known principles of statistical model equivalence.

- **Structural Operators over Γ_Y :** The user introduces a *neutral operator* $\mathcal{N}(\Gamma_Y, D_f)$ defined “purely at the structural level” of Γ_Y . This appears to be an operator that encapsulates the one-step dynamics encoded in Γ_Y *without reference to a specific state-space realization*. If Y were itself a Markov process, $\Gamma_Y(dy, dy')$ is essentially the Markov transition kernel, which can be seen as an operator $P: L^2(Y) \rightarrow L^2(Y)$ (or an operator on measures) via $(Pf)(y) = \int f(y') \Gamma_Y(dy' | y)$. In general Y might not be Markov on its own (e.g. Y could be a scalar observation of a higher-dimensional system, so the sequence of Y ’s is an HMM with memory). Nevertheless, any joint distribution on consecutive pairs defines a *one-step transfer operator* on functions of Y (though it may not satisfy the Chapman-Kolmogorov consistency beyond one step). We did not find existing literature on an operator that is *defined directly from empirical two-point data and independent of any chosen state-space*. This “neutral” operator seems to be a novel construct. It effectively treats Γ_Y itself as an object that can be factorized or manipulated: all candidate models are seen as different factorizations of Γ_Y (either via a latent variable X or via a mapping $r: Y \rightarrow Z$ to a representation space). In classical terms, Γ_Y is like the *empirical transition matrix* for the observed process, and one can ask: does there exist an actual Markov chain on some latent X that has observation kernel $h: X \rightarrow Y$ yielding exactly Γ_Y ? This is reminiscent of the filtering or predictive state representations literature (where one uses observable transitions to construct a state). Indeed, work on **Predictive State Representations (PSRs)** in machine learning showed that one can construct a state vector out of observable futures such that the system becomes Markov in that predictive space ³⁵. The “neutral operator” might be interpreted as the *observable-state transition operator* acting on such predictive states of the process. However, framing it as $\mathcal{N}(\Gamma_Y, D_f)$ and emphasizing its *divergence to any specific model’s operator* is new. In summary, treating Γ_Y itself as an operator (perhaps a *Perron-Frobenius operator* on observed measure space) that is “neutral” among all realizations is a novel idea, albeit inspired by the standard concept of a transition kernel.

- **Semigroups and IG Foliations with Spectral Theorems:** The program envisions a *dynamical neutral operator* $\mathcal{N}_{\text{dyn}}(\Gamma_Y, D_f)$ that carries a semigroup structure (time-evolution) and studies its spectral properties on *IG leaves*. In classical spectral theory, if two dynamical systems are *conjugate* (in the topological or measure-preserving sense), their Koopman and Perron-Frobenius operators are conjugate (via unitary or similarity transforms) and hence share spectra. Here, however, the context is different: two models in the same “IG leaf” might not be strictly dynamically conjugate in the usual sense, yet the claim is that within each IG-equivalence class (perhaps defined by the constraint Γ_Y and other distributional features), there are *conjugacy theorems for resolvents and generators*. We did not find existing theorems along the lines of “if two systems lie in the same IG leaf, their Koopman generators are spectrally conjugate.” This appears to be a *conjecture or result proposed by the user*. In linear systems theory, a related fact is that two different realizations of the same input-output behavior have transfer functions with identical pole spectra (eigenvalues of the state transition matrix that are “observable”) – that’s a control theory analog. For nonlinear systems, there is no general result beyond special cases (e.g. topological conjugacy preserves *all* spectral invariants by definition, but IG-equivalence is weaker than

conjugacy). The mention of *resolvent spectra* suggests using functional analysis: the resolvent $(\lambda I - A)^{-1}$ of a generator A and its spectrum. Possibly the user proves that if two models produce the same one-step distribution (and maybe share certain higher-order distributions), one can construct an intertwining mapping between their associated operators restricted to certain invariant subspaces (like the span of observables in the σ -algebra of Y) so that the nonzero spectral values coincide. This would be a novel spectral theorem, as we did not encounter it in literature. Prior work that comes somewhat close might be in *ergodic theory* and *functional equivalences*: e.g., isospectral flows or lumped Markov chains (where different Markov chains can have the same eigenvalues if their transition graphs are related by certain quotients). But the explicit combination “semigroup + IG foliation” with spectral conjugacy is likely a **genuinely novel** contribution of this framework.

- **Spectrally Correct Embedding Dimensions (IG + Operator theory):** Selecting an embedding dimension that is “spectrally correct” means choosing the dimension of a reconstruction (e.g. number of delays, or latent state dimension) such that the important spectral features (eigenvalues, continuous spectrum components) of the dynamics are captured. Traditionally, Takens’ theorem gives a *topologically* correct embedding dimension (to capture the attractor geometry but not quantitative spectral details), while more recent Koopman-based heuristics give a *practical* embedding length (e.g. using delays until singular values of the Hankel matrix drop off, as in time-lagged DMD methods ³⁶–³⁷). The user’s approach of defining this via *IG immersion and resolvent conjugacy* is new. It implies one immerses the system’s statistical manifold into a larger one (by adding dimensions or features) until the *resolvent spectrum* of the neutral operator stabilizes (or matches that of a hypothetical true system). We did not find prior literature that connects information geometry immersions with the spectral accuracy of embeddings. However, conceptually it resonates with two known ideas: (1) *Observability*: In linear systems, the minimal embedding dimension to capture the system’s eigenvalues is the observability index – if a mode is unobservable, no finite embedding of outputs will see it. (2) *Delay embedding for Koopman*: Arbabi and Mezić (2017) showed that as one increases the number of delays, the approximated Koopman spectrum converges to the true spectrum for an ergodic system ¹⁶. But formally choosing a finite “spectrally sufficient” number of delays is not solved in general. The user’s contribution seems to be defining this rigorously via an IG criterion (perhaps requiring that the *Fisher information immersion* of the model into the larger space is non-degenerate and that the model’s generator is conjugate to the neutral operator’s generator in that immersion). This is a **nontrivial generalization** of known embedding theorems, introducing spectral and information criteria rather than just topological ones. Thus, the idea of *spectrally correct embedding dimension* that unifies Takens embeddings with operator spectra appears to be novel.

- **Empirical Separability via Block-Diagonal IG Structure:** This refers to detecting when the observed/latent/representation spaces can be “cut” such that their influences separate, manifesting as a block-diagonal Fisher information metric (and corresponding dual affine connections). In classical IG, a block-diagonal Fisher information matrix indicates that the parameter can be split into independent subsets that do not interact to first order (no mixed partial derivatives of the log-likelihood). In model terms, this often corresponds to statistical independence or a product structure in the joint distribution. For example, if a model’s latent and observed components were independent, the Fisher information would factor into two blocks. In the context of dynamical systems, “empirical separability across cuts” might mean that one can partition the state into modules (or separate a representation into independent factors) such that each module’s

parameters can be estimated independently (no Fisher coupling) and each module evolves independently at the level of information. We did not find a prior framework articulating this for dynamical systems. However, the concept draws on known ideas: e.g., in an HMM with independent emission and transition parameters, the Fisher information matrix is block-diagonal between those parameter sets. More generally, Amari’s theory of *multivariate information orthogonalization* might be relevant – he discussed decomposing interactions into orthogonal components ³⁸. The user’s notion of using *excess Fisher curvature* as a measure of irreducible coupling is new. It implies that if the Fisher information metric (plus its dual connections) can be transformed to block-diagonal form by some choice of coordinates (or representation), then the system is *empirically separable* into mostly independent parts; if not, the residual cross terms (curvature) quantify the coupling that no reparameterization can eliminate ³⁹ ⁴⁰. No existing literature was found that formulates “separability” in exactly this way, so this is likely a novel diagnostic tool. It parallels concepts in representation learning (where one seeks statistically independent latent factors) but from an IG angle. In summary, *block-diagonal Fisher structure = independent subprocesses* is well-understood; using that as an **empirical test** on learned representations to validate if latent factors have been disentangled is a new application of IG.

Mapping of Constructs to Known Frameworks

To clarify how the user’s constructs relate to existing concepts, the following table provides a mapping. Each item of the proposed framework is matched to analogous ideas in the literature and categorized as a rephrasing of known ideas, a nontrivial generalization, or a novel concept:

User’s Construct	Related Known Frameworks	Relation/Novelty
Empirical edge marginal $\Gamma_{Y \in P(Y \times Y)}$ – one-step joint distribution of observations.	<i>Transition probability</i> in a Markov chain; empirical two-point distribution of a time series.	Rephrasing: It’s a standard object (the empirical joint measure of consecutive observations) ⁴¹ , commonly used to estimate Markov transitions or autocorrelations. The novelty is treating it as the fundamental “operator” object for IG analysis.
Realization space $\mathcal{R}(\Gamma_Y)$ of latent operators $(X, \Gamma_{X,h})$ etc. with pushforward π to Γ_Y .	<i>Hidden Markov model (HMM) equivalence class; system identification’s family of realizations.</i>	Mostly known concept: All state-space models yielding the same observable joint distribution form an equivalence class ³⁰ . Prior results (e.g. minimal HMM uniqueness ³¹) exist, but using IG and divergence measures to characterize this space is a new angle.
IG structure from f-divergences and α-divergences (Fisher metric, dual affine connections).	<i>Information geometry of statistical manifolds (Amari 1985, 2016); Fisher–Rao metric, α-connections from Csiszár divergences.</i>	Established framework: This is classical information geometry ¹ . The user is applying these known IG tools to dynamical-system-specific manifolds (which is innovative), but the IG ingredients themselves are standard.

User's Construct	Related Known Frameworks	Relation/Novelty
IG leaves via conditional-law maps Φ_S and Fisher immersions.	<i>Fibers in statistical manifolds defined by fixing certain conditional distributions; hierarchical modeling.</i>	Generalization of known ideas: Amari's work on hierarchical models and Markov chains provides analogous constructs (e.g. fixing lower-order marginals defines a submanifold) ²⁵ . The "leaves" here resemble fibers of a fibration where each leaf = all distributions sharing some conditional structure. Using Fisher immersions (inclusion maps of these leaves into a larger manifold) is a natural IG concept, but applying it to dynamical realizations is new.
Neutral operator $\mathcal{N}(\Gamma_Y, D_f)$ (structural-level operator derived from Γ_Y alone).	<i>Markov chain transition operator on Y (if Y were Markov); observable operator in predictive state representation.</i>	Novel concept: No prior named "neutral operator." It rephrases the idea of an abstract operator that any realization must emulate on observables. It draws from the Markov transition matrix idea and PSR/observable operator models, but explicitly defining it via an f-divergence criterion (neutral w.r.t. model choice) is new.
Dynamical neutral operator $\mathcal{N}_{\text{dyn}}(\Gamma_Y, D_f)$ with semigroup dynamics.	<i>*Semigroup of Markov operators on measure space; shift operator on sequence space of Y.</i>	Novel extension: If Y -process is not Markov of order 1, a one-step $\mathcal{N}_Y$ doesn't directly yield a semigroup for multi-step transitions. The framework likely extends $\mathcal{N}$ to a consistent family $\{\mathcal{N}^t: t=1,2,\dots\}$ that matches empirical t -step marginals. This idea of an empirical semigroup on the space of Y -distributions is new, though it relates to constructing a <i>Markov approximation</i> or using empirical transition matrices iteratively.
Conjugacy of resolvent spectra and generators within IG leaves.	<i>Unitary or similarity conjugacy of Koopman/Perron-Frobenius operators for conjugate systems; spectral invariants under model equivalence.</i>	Likely novel theorem: In linear systems, equivalent realizations yield the same transfer function poles (observable spectrum), and isomorphic Markov chains share spectrum. But a general statement for nonlinear systems in the same "IG class" is new. It suggests a deep result that all models reproducing the same Γ_Y (and perhaps higher-order stats) share some spectral properties (perhaps via an intertwining operator). This goes beyond existing spectral theory.

User's Construct	Related Known Frameworks	Relation/Novelty
Spectrally correct embedding dimension (via IG immersion + resolvent conjugacy).	<i>Takens embedding dimension for topological equivalence; observability index in linear systems; Hankel-DMD convergence for Koopman spectra.</i>	New integration of ideas: Determining embedding dimension by requiring preservation of spectral features is a novel criterion. Past methods are heuristic or topological ⁴¹ or linear-specific. Using an IG immersion (ensuring the Fisher information captures the needed degrees of freedom) combined with a spectral matching condition is an innovative generalization on how to choose embeddings. Delay embeddings appear as a special case (embedding in time-lag coordinates), but the framework could encompass learned embeddings as well – an expansion of known theory.
Empirical separability across cuts (block-diagonal Fisher metric, dual connections).	<i>Decomposable models with independent components; product manifold structure yielding block-diagonal information matrix.</i>	New diagnostic usage: The fact that statistical independence or orthogonal parameterization leads to block-diagonal Fisher information is well-known. For example, in a mixture of independent submodels, the Fisher metric splits into blocks. The novelty is using this as a criterion on empirical data: if a learned representation Z yields a Fisher information matrix that is (approximately) block-diagonal between (Y, Z) or between latent factors, one can declare those factors empirically disentangled. This takes a classical IG notion (block-diagonal = no coupling) and applies it in a new way to judge model sufficiency or factorization in dynamical settings.

Novelty and Likely Contributions of the Framework

Overall, the proposed *neutral operator framework* appears to be a significant synthesis of ideas from information geometry, dynamical systems, and operator theory. Many of its building blocks are drawn from established domains, but their combination yields several **novel insights and tools**. Key likely contributions and novelties include:

- **Unifying Representation and Dynamics via an “Observable Operator”:** By centering the analysis on the empirical γ_Y (the joint distribution of successive observations) and viewing every model as just a different factorization of this object, the framework provides a *coordinate-free, representation-agnostic* viewpoint. This is philosophically appealing – it treats the data (observable transitions) as primary and models as secondary. While similar sentiments exist in predictive state representations and in Čencov’s invariant geometry (the idea of studying objects that don’t depend on particular parametrization³⁴), here it is operationalized in a dynamical context. This could greatly clarify comparisons between, say, a delay embedding and a hidden Markov model: both

become “alternative factorizations” of the same Γ_Y ⁴² ⁴³. Such a unification of reconstruction (Takens embedding), inference (HMM/PSR models), and representation learning (autoencoders that impose a mapping $r: Y \rightarrow Z$) under one umbrella is a novel conceptual contribution.

- **Hierarchy of Fidelity Criteria (Topological, Measure, Geometric):** The program delineates multiple levels of model fidelity to data: (1) *Topological fidelity* (as in Takens’ theorem – does the model embed the true dynamics diffeomorphically?), (2) *Measure fidelity* (the “Rose” condition in the user’s abstract – requiring $D_f(\Gamma_Y|\Gamma_Z)=0$ for the model Z , meaning the model reproduces one-step transition statistics exactly) ⁴³ ⁴⁴, and (3) *Geometric fidelity* (requiring Fisher information additivity, i.e. the model doesn’t introduce spurious information curvature across chosen cuts) ³⁹ ⁴⁰. While each of these conditions individually has appeared in literature (Takens for topology, maximum-likelihood or moment matching for distribution, Fisher metric checking for lack of misspecification), **articulating them as a hierarchy of increasingly stringent criteria is new**. It provides a language to discuss model adequacy: for example, a delay embedding might achieve topological fidelity but not measure-fidelity (it can capture the shape of attractor but not exact statistical distribution unless infinitely long delays are used), whereas a well-tuned HMM might achieve measure fidelity but perhaps not geometric fidelity if it still fails to capture some higher-order dependencies (manifested as curvature). This stratified view is an original contribution that could clarify discussions of model quality in system identification.
- **Introduction of Excess Fisher Curvature as a Diagnostic:** In IG, curvature of the dual connections measures the departure from exponential family structure. The user’s notion of “excess Fisher curvature” quantifies *coupling that no change of coordinates can remove* ³⁹. If a model perfectly explains the data’s dependencies, one would expect that in the right coordinate system (model’s natural parameters) the Fisher information would factor and curvature would vanish (the model lies on a flat submanifold of the true distribution manifold). If not, the remaining curvature signals model mismatch or latent coupling. This is a novel diagnostic tool. It goes beyond classical goodness-of-fit tests by providing a differential-geometric measure of *where* and *how* a model fails. For instance, if an assumed factorization of dynamics (like treating subsystems as independent) is incorrect, the Fisher information will have off-diagonal terms and nonzero α -connection curvature even if the one-step distributions fit perfectly. This use of IG curvature to assess dynamical model adequacy is a fresh idea that bridges theory (curvature) with practice (model error).
- **Spectral-IG Correspondence for Model Equivalence:** Perhaps the most theoretically deep contribution is the claim (or conjecture) that models in the same IG leaf share spectral properties via resolvent conjugacy. If proven, this is a powerful result: it means that from purely observational data (distributional equivalence), one can infer non-trivial spectral invariants of any realization. Practically, this could justify computing spectra on a delay-embedded observable and claiming they are intrinsic to the underlying system (not an artifact of the particular model chosen). This idea hasn’t been rigorously present in literature – it’s an intriguing new bridge between *statistical indistinguishability* and *spectral (dynamical) indistinguishability*. It could answer questions like: do two different minimal realizations of a process necessarily have the same Koopman eigenvalues (even if their state variables differ)? The framework leans toward yes, within each information-equivalence class. If established, that is a significant theoretical contribution.

- **“Spectrally Correct” Model Reduction and Embedding:** By combining IG and operator theory, the framework suggests a method for model reduction: increase the dimension of your representation until the *spectral invariants* stop changing and the *information curvature* vanishes beyond a certain point. This is akin to finding the minimal sufficient latent dimension that can capture the system’s dynamics. Classical nonlinear dynamics lacked a principled criterion for the minimal embedding dimension aside from “no false neighbors” heuristics or empirical saturation of invariants. The proposed use of resolvent (eigen-spectrum) convergence as a criterion, together with IG measures to ensure statistical consistency, is new. It could give a more objective way to choose dimension in algorithms like delay-coordinate regression or autoencoder-based embeddings (e.g., one could embed in successively higher dimensions until the divergence between Γ_Y and Γ_Z is below a threshold and the estimated spectrum of $N_{\text{cal}}^{\text{dyn}}$ stabilizes). This *marriage of quantitative spectral analysis with statistical model selection* appears to be unique to this framework.

In summary, the framework’s novelty lies not in inventing brand-new mathematical machinery, but in **integrating multiple perspectives into a coherent theory and extending them**. It stands to contribute: a unified perspective on modeling dynamical systems (treating all models equally as parameterizations of data), new theoretical results linking statistical equivalence to spectral equivalence, and practical criteria (based on divergences and information metrics) for tasks like determining embedding dimension and detecting when a representation has fully captured the system. This level of integration of IG with dynamical systems/semigroup theory is unprecedented in the literature reviewed.

Relevant Research Communities

The work is highly interdisciplinary. Several research communities would find this framework of interest, and framing it appropriately will be important for legibility:

- **Information Geometry & Statistical Inference Community:** Researchers focused on information geometry (e.g. those who attend the “Geometric Science of Information (GSI)” conferences or publish in the *Information Geometry* journal) would be naturally interested in this work. They will appreciate the novel use of α -divergences, Fisher metrics, and dual connections in a dynamical setting. The framework’s emphasis on invariants under reparameterization and the use of Csiszár divergences aligns well with the IG community’s values ¹. By highlighting the new geometric results (like curvature diagnostics and IG leaves foliation), one can engage this community.
- **Dynamical Systems and Ergodic Theory Community:** Applied dynamical systems theorists (including those working on chaos, strange attractors, and time-series analysis) would be another audience. In particular, the Koopman operator and nonlinear dynamics community – which often publishes in venues like *SIAM Journal on Applied Dynamical Systems* or *Chaos* – has been seeking systematic ways to connect data-driven models to true dynamics ⁴⁵ ⁴⁶. They would be interested in the spectral aspects and the conditions for faithful embeddings. The notion of “topological vs measure vs geometric fidelity” could resonate with dynamical systems people who distinguish between topological conjugacy, measure equivalence, etc. Emphasizing how this framework helps choose embedding dimensions or detect hidden Markov structure in dynamics will catch their attention.

- **Control Theory and System Identification:** The control/identification community (e.g. those at the IEEE Conference on Decision and Control, or journals like *Automatica* and *IEEE Transactions on Automatic Control*) value results on minimal realizations, state-space factorization, and observability. This work recasts the realization problem in a fresh way (using divergences instead of just algebraic equivalences) ³⁰, and introduces spectral tests for model reduction. By framing parts of the work as solving an identification problem (how to go from Γ_Y to a minimal (X, Γ_X, h)) or as providing guarantees for model reduction, one can interest this community. The mathematical style (using operator semigroups and functional equations) is also something control theorists are comfortable with. Care should be taken to clarify terms (e.g. translating “Fisher metric block-diagonal” to “unbiased decoupled estimation of parameters” in control terms) for better reception.
- **Machine Learning and Time-Series Modeling:** Given the surge in data-driven modeling of dynamical systems (e.g. recurrent neural networks, variational autoencoders for dynamics, reinforcement learning), parts of this framework could appeal to the theoretical machine learning community. In particular, those working on **predictive state representations (PSRs)** and **observable operator models** in ML would recognize the theme of using only observable quantities to represent state ³⁵. The IG approach provides new tools (e.g. using KL-divergence as a training objective and Fisher information to assess representation quality) that could augment machine learning approaches. Additionally, the idea of encouraging block-diagonal Fisher information for disentangled representations touches on unsupervised representation learning (a topic at ICML, NeurIPS, etc.). While the work is quite theoretical, if some empirical demonstrations are provided (e.g. learning a latent representation that achieves geometric fidelity and showing it improves forecasting or control), ML venues might be interested. At a minimum, workshops or special sessions on “Learning theory for dynamical systems” could be suitable.

In summary, **Information Geometers, Dynamical Systems theorists, Control/Systems theorists, and theoretical ML researchers in sequence modeling** all form potential audiences. The common thread is an interest in rigorous frameworks for understanding complex dynamic processes; this work offers each something novel (be it a new geometric invariant or a new identification criterion).

Suitable Journals and Conferences (Theory-Heavy Work)

Given the theoretical nature of the contribution, one would target journals and conferences that value foundational advances in modeling and analysis:

- **Journal of Applied Dynamical Systems (SIAM JADS)** – A top journal for dynamical systems theory and algorithms. It has published Koopman operator works and could handle the mix of operator theory and geometry.
- **Chaos (AIP)** – An interdisciplinary journal on nonlinear science. It welcomes innovative methods for analyzing chaotic time series and dynamics. Prior works on Koopman analysis and information-theoretic measures in dynamics have appeared here, making it a good fit ¹⁰ ⁴⁷.
- **Physica D: Nonlinear Phenomena** – A well-regarded journal for theoretical developments in nonlinear dynamics. It has historically published information geometry approaches to dynamics (e.g. geometrical structure of gradient flows ⁴⁸) and could be appropriate for a lengthier theory paper.

- **Information Geometry (Springer)** – A newer journal specifically focusing on IG. This work squarely lies in that scope, applying IG to a novel domain. The journal (and its community) would appreciate the depth on divergences, Fisher metrics, and might allow a more expansive discussion of philosophical underpinnings ⁴⁹.
- **IEEE Transactions on Automatic Control** – If the paper is framed with a bit more systems/control theory terminology (focusing on realizations, minimality, etc.), TAC could be an option. It requires a strong theoretical contribution. The spectral and divergence-based criteria for model equivalence could be pitched as a new systems theory result. (Automatica or SIAM Journal on Control and Optimization are similar alternatives in this vein.)
- **SIAM Journal on Mathematics of Data Science (SIMODS)** – This journal might fit if the work is presented as a method for data-driven modeling of dynamics. The mix of geometry and spectral analysis for learning models aligns with the “math of data” theme. It’s a more applied choice compared to pure IG or DS journals.
- **Entropy (MDPI)** – This journal is devoted to information-theoretic measures and often features interdisciplinary work mixing information measures with physics and engineering. A paper focusing on the divergence aspects (Csiszár–Amari divergences, Fisher information as entropy curvature) ⁴⁴ ¹ could find a home here. (MDPI journals are less selective, but they actively seek IG-related content.)
- **NeurIPS (Conference) or ICLR (Conference)** – For reaching a machine learning audience, these top conferences could be considered *if* the work includes empirical validation on, say, learning a dynamical system representation using the proposed criteria. The core ideas are theoretical, but NeurIPS has a track for theoretical ML. Emphasizing how this framework can guide representation learning (perhaps an algorithm that minimizes the $D_f(\Gamma_Y|\Gamma_Z)$ divergence subject to Fisher block-diagonality) would be key. The bar for clarity and empirical support is high, however.
- **Geometric Science of Information (GSI) – Conference** – This biennial conference (with Springer proceedings) is tailored to information geometry applications. A paper here would reach the IG community directly. The review threshold might be more approachable for a theory-heavy piece bridging IG and dynamics, and it would ensure the work is visible to experts in IG.
- **IEEE Conference on Decision and Control (CDC)** – A premier conference for systems and control theory. If the paper is angled towards system identification and observer design (e.g. constructing an observer via neutral operator, or conditions for minimal sufficient statistics in control), CDC could be a venue. It requires a succinct format, so perhaps a focused subset of results would be presented, with the full details in a journal.

Given the breadth of the work, it might even spawn multiple publications: one in an IG-focused venue laying out the theoretical foundations (divergences, geometry, conjugacy theorems), and another in a dynamics/control venue emphasizing spectral analysis and model construction. **In summary, journals like *Information Geometry*, *SIAM J. Applied Dyn. Sys.*, *Chaos*, *Physica D*, or *IEEE TAC*, and conferences like *NeurIPS* (for theoretical ML), *GSI* (for IG), or *CDC* (for control) are prime candidates.**

Follow-Up Research Directions

To build on this framework and strengthen its impact, a number of follow-up directions can be pursued:

- **1. Rigorous Theoretical Proofs and Generalizations:** Several claims (e.g., resolvent spectral conjugacy within IG leaves, existence and uniqueness of the neutral semigroup) would benefit from formal proofs. A theoretical extension could be to prove *under what conditions on Γ_Y and divergence D_f the neutral operator $\mathcal{N}$ generates a well-defined C_0 -semigroup*. This might involve showing that the empirical one-step operator is *quasi-compact* or satisfies Hille–Yosida conditions ⁵⁰. Another theoretical direction is to generalize the conjugacy theorems: for instance, establish that if two dynamical systems share not just Γ_Y but a finite history of length m (i.e. their m -step joint distributions agree), then their Koopman operators have the same eigenvalues up to a certain index (related to the observable Markov order). This would formalize the intuition of spectral equivalence as more information is shared.
- **2. Algorithm for Model Construction from Data (Estimation):** The framework outlines a pipeline (estimate Γ_Y , then find a model Γ_Z minimizing divergence, etc.) ⁵¹ ⁵². Operationalizing this into an algorithm is an important next step. One could develop an optimization procedure to find a realization (for example, an HMM or a parametric nonlinear model) that best matches the empirical Γ_Y in D_f divergence. This is akin to the *inverse problem* of model fitting, but here framed as minimizing an f -divergence between empirical and model one-step transitions. A practical follow-up is to examine the computational aspects: e.g., using gradient descent or natural gradient (Amari’s natural gradient method ⁵³) on the space of model parameters to reduce $D_f(\Gamma_Y | \Gamma_Z)$. Additionally, one could incorporate regularization to avoid overfitting (since empirical Γ_Y is noisy). This yields a data-driven algorithm for system identification grounded in IG – testing it on synthetic and real datasets would strengthen the contribution.
- **3. Empirical Validation on Complex Systems:** To demonstrate the power of the neutral operator approach, applying it to a few benchmark dynamical systems would be valuable. For example, one could take a known chaotic system (say the Lorenz attractor), generate observation data, estimate Γ_Y , and then use the framework to discover a minimal embedding dimension. Compare the spectrally determined embedding dimension to the known ground truth (Takens’ $2d+1$) and to other methods (false nearest neighbors, Hankel-DMD). Similarly, one could test an HMM example: generate sequences from a hidden Markov process, estimate Γ_Y , and see if the divergence-minimization recovers the correct number of hidden states. Empirical results showing that *Fisher curvature goes to zero* precisely when the model complexity reaches the true complexity would strongly support the framework’s validity. This could also uncover any gaps (e.g., small sample issues, the need for higher-order marginals beyond Γ_Y).
- **4. Extension to Continuous-Time and Non-Stationary Systems:** The current framework seems oriented to discrete-time, stationary processes. A theoretical extension would be to continuous-time dynamics (flows). In that case, Γ_Y might be replaced by an *infinitesimal generator* (estimated from data via transition densities at small Δt). One could define a “neutral generator” $\mathcal{N}_{\mathrm{dyn}}$ and require divergence-based equivalence for *infinitesimal* transitions. This would connect to the theory of Markov processes and their Fisher information geometry (as developed in e.g. thermodynamic systems ²⁹). Non-stationary or time-

varying processes could also be considered: then Γ_Y becomes time-indexed $\Gamma_Y(t, t+\tau)$. Extending the neutral operator to handle slowly varying dynamics or periodic drives (perhaps by augmenting state with time) would broaden applicability. These extensions would involve significant new theory to handle time dependence in the IG setting.

- 5. Integration with Optimal Transport and Causal Inference:** The user’s framework focuses on f -divergences (like KL). Another rich perspective comes from *optimal transport metrics* (Wasserstein distances) on probability distributions, which have their own geometry. A follow-up direction could be to compare and integrate the IG approach with optimal transport: for instance, does minimizing a KL divergence between Γ_Y and Γ_Z yield similar model alignment as minimizing a Wasserstein distance between their distributions? Recent work by Amari and others has looked at connecting Wasserstein geometry with information geometry ⁵⁴. Bringing that into dynamical systems (perhaps defining a “neutral operator” in the OT sense) could lead to hybrid criteria that account for both information and state-space distance. Additionally, investigating *causality*: since Γ_Y encodes forward-in-time influence, one might ask if the neutral operator framework can help infer causal structure (for example, in a multivariate system, block-diagonal Fisher information could indicate lack of causal links between variables). This would align with the growing interest in causal discovery from time series.
- 6. Application to Representation Learning (in ML or Neuroscience):** The block-diagonal Fisher information idea directly suggests an application: one could design representation learning objectives that encourage the Fisher information matrix of (Y, Z) to be approximately block-diagonal, thus yielding “separated” latent factors. A concrete follow-up is to implement a neural network or autoencoder that, given time-series data Y , learns a mapping $r: Y \rightarrow Z$ by optimizing a loss that is a weighted sum of: reconstruction of Γ_Y (e.g. via a likelihood or divergence term) and a term penalizing off-diagonal Fisher information between Y and Z factors. This would force the representation Z to capture all predictive information about Y (to match Γ_Y) but in a form that is as disentangled as possible (no excess coupling). Such an approach could be tested on datasets where the ground-truth latent factors are known (e.g., two independent oscillators observed in a mixed way). If the method successfully learns the correct factors, it would demonstrate the practical value of the theoretical metrics. This could appeal to both ML practitioners and neuroscientists (who might use it to find independent components driving neural recordings, for example).
- 7. Connection to Reinforcement Learning and Control:** Yet another direction is to apply the neutral operator concept to *predictive modeling for control*. In RL, one often needs a model of state transitions (the “dynamics model”) and of observations. The divergence-based neutrality condition $D_f(\Gamma_Y | \Gamma_Z) = 0$ could be used as a constraint to learn a model that is as good as the real system in one-step prediction. Similarly, Fisher geometry could be used to assess if the learned model has redundant degrees of freedom. A practical follow-up might be: train two models of a dynamical system (one large neural network, one smaller network) and use the framework’s invariants to decide if the smaller model has captured all necessary dynamics (if yes, Γ_Y distributions align and Fisher curvature is low). This would provide a systematic model compression tool for control. Integrating such IG-based tests into control system identification (for example, in subspace identification algorithms, use D_f to choose model order) could significantly improve reliability of learned controllers.

- **8. Deeper Analysis of Divergence Choice (Role of α):** The framework mentions that different α -divergences (forward KL, reverse KL, symmetric, etc.) put emphasis on different aspects of model-data mismatch ⁴⁴ ⁵⁵. A theoretical follow-up is to analyze how the choice of divergence affects the neutral operator and the resulting model. For example, forward KL (data || model) will prioritize explaining the empirical transitions (avoiding “unexplained” data mass) ⁴⁴, whereas reverse KL (model || data) avoids model predictions that don’t appear in data ⁵⁵. One could derive conditions under which these divergences lead to the same neutral operator (the user notes they share the zero set and induce the same Fisher metric ⁵⁶). It would be insightful to explore situations where they differ – perhaps in undersampled regimes or if the model class is misspecified. This analysis could guide practitioners on which divergence to use as the “edge action” cost for a given task (e.g. if false negatives vs false positives in transitions have different penalties).
- **9. Higher-Order Extensions (beyond one-step):** The current neutral operator focuses on one-step transitions. A natural extension is to consider Γ_{Y^k} , the joint distribution of (Y_t, Y_{t+k}) for some lag k . By increasing k , one incorporates more long-term dynamics. One follow-up is to develop a *multi-step neutral operator* or a set of operators for a range of lags, and define neutrality in terms of matching all those distributions. This relates to the concept of *moment closure* or *Kolmogorov extension*: ensuring a candidate model is consistent with multi-time marginals. It might enhance the framework’s ability to capture systems with longer memory. Theoretically, this could be framed as an IG foliation where each leaf is defined by the entire *process* law up to some order. Investigating the trade-off between k (the memory length considered) and model complexity would extend the utility of the approach (e.g., is a slightly larger state space that exactly matches two-step statistics preferable to a minimal state that only matches one-step?). This also ties to the idea of spectral content: matching higher-order marginals should enforce correctness of not just eigenvalues but also pseudoeigenvalues (resonances) associated with longer-time correlations.
- **10. Bridging to Quantum or Non-commutative Settings:** For a more theoretical/mathematical follow-up, one could ask if the neutral operator concept extends to *quantum dynamical systems*, where Γ_Y might be replaced by a density matrix or a completely positive map between operator algebras. Information geometry has quantum analogues (monotone metrics, Chentsov’s theorem in quantum context) ⁵⁷. Dynamical systems have Koopman operator analogues in unitary evolution on Hilbert spaces. If one could define a “neutral quantum channel” for a given observed process and use quantum f -divergences, that would open a new frontier (though admittedly abstract). It would connect this framework to the burgeoning field of quantum information geometry. Even if not quantum, one could consider *non-commutative probability models* (e.g. operator-valued measures) – potentially relevant if the observed data has algebraic constraints.

Each of these directions leverages a different aspect of the framework – some solidify its theoretical foundations, others demonstrate its practical applicability, and others broaden its scope. Pursuing a few of them would greatly strengthen the original contribution, making it not only a standalone theory but a springboard for diverse future research.

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